19 OCT 2026

25AIMLPCC304ESE

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

End Semester Examinations, October 2026

B.Tech. Artificial Intelligence and Machine Learning - SEMESTER -III

25AIMLPCC304 - Database Management System

Time : 2.00 Hours **Maximum Marks : 60**

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

St. John College of Engineering & Management SJCEM PALGHAR

Section I Maximum marks (12)

Q1	Attempt all Questions	Mrk	Course Outcome	RBT Level
a)	What is data independence? Explain its types.	2	CO1	L2
b)	Explain the selection operation in relational algebra.	2	CO2	L2
c)	Write an SQL command to create a Student table with suitable attributes.	2	CO3	L3
d)	Convert a relation containing repeating groups into 1NF.	2	CO4	L2
e)	Apply a relational algebra transformation to simplify a query.	2	CO5	L3
f)	List different editions of IBM Db2.	2	CO6	L2

Section II Maximum marks (30)

Q2	Attempt all Questions	Mrk	Course Outcome	RBT Level
i	a)Analyze the relationship types (one-to-one, one-to-many, and many-to-many) in a university database and justify their use.	5	CO1	L4
	OR			
	b)Analyze the impact of the absence of data independence in a database system. Support your answer with a suitable example.	5	CO1	L4
ii	a)A database contains Supplier and Product relations. Apply the Cartesian Product operation and explain how it can be used as a preliminary step before applying a Join operation.	5	CO2	L3

	OR			
	b)Explain how Union and Intersection operations can be used together to solve database retrieval problems with suitable examples.	5	CO2	L3
iii	a)Write a nested SQL query to retrieve the names of employees earning more than the average salary.	5	CO3	L3
	OR			
	b)Write SQL statements using GRANT and REVOKE commands to provide and remove SELECT and INSERT privileges for a user.	5	CO3	L3
iv.	a)Differentiate between 3NF and BCNF based on their conditions, determinants, functional dependencies, and decomposition requirements.	5	CO4	L4
	OR			
	b)Compare decomposition using functional dependencies with decomposition using multivalued dependencies. Explain when each type of decomposition is required.	5	CO4	L4
v	a)Given two transactions with their read sets and write sets, analyze whether the transactions can be validated successfully	5	CO5	L4
	OR			
	b)After a successful bank transaction, the server crashes and restarts. Analyze which ACID property ensures that the committed transaction is not lost.	5	CO5	L4
vi	a)Analyze the factors that should be considered when selecting SQL Server over another RDBMS for an enterprise application.	5	CO6	L3
	OR			
	b)Explain the basic Db2 CLP commands used for database connection, SQL execution, and database disconnection.	5	CO6	L3

Section III Maximum marks (18)

Q3	Attempt all Questions	Mrk	Course Outcome	RBT Level
i	Explain the difference between an entity type and an entity set with a suitable example.	3	CO1	L2
ii	Explain the basic components of a tuple relational calculus expression.	3	CO2	L2
iii	Explain COUNT, SUM, AVG, MIN, and MAX functions with suitable examples.	3	CO3	L4
iv	Analyze a set of functional dependencies and identify the attributes that determine other attributes.	3	CO4	L2
v	Explain the concept of serial and non-serial schedules.	3	CO5	L4
vi	Analyze a database requirement involving constraints and NULL values and determine suitable Db2 SQL operations.	3	CO6	L4